Classification of AI Environments

The two systems from Task 1 are the Delivery Robot and the LLM-Based Student-Support Agent. Their environments are classified according to observability and the predictability of action outcomes.

1. Delivery Robot

Dimension	Classification	Justification and example
Observability	Partially observable	The robot's sensors cannot detect everything in its surroundings. For example, a person hidden behind a corner may not be visible to its camera or LiDAR sensors.
Determinism	Stochastic	Action outcomes are uncertain because pedestrians may move unexpectedly, wheels may slip, or obstacles may appear suddenly. For example, the robot's planned movement may be interrupted by a pedestrian crossing its path.

Operational assumption: The robot operates on a university campus with cameras, distance sensors, and a digital map. Its real environment is partially observable and stochastic because sensor coverage is limited and external events are unpredictable.

2. LLM-Based Student-Support Agent

Dimension	Classification	Justification and example
Observability	Partially observable	The agent can read student messages and approved course materials, but it may not know the student's full circumstances or whether a document is outdated. For example, a student's unreported deadline extension may be hidden from the agent.

Determinism	Stochastic	An LLM may generate different answers to the same question because of sampling, model settings, or context differences. Tool results may also change when university information is updated. For example, the agent may produce different explanations for the same assignment question.

Operational assumption: The agent can access the student-support chat, approved course documents, read-only LMS search, and a support-ticket creation tool. It cannot access private grades or change student records without authorization.

3. Real Environment vs. Simplified Simulator

System	Real environment	Simplified simulator
Delivery robot	Partially observable and stochastic due to hidden obstacles, sensor limitations, and unpredictable pedestrian movement.	May be fully observable and deterministic if the simulator displays every object and applies fixed movement rules.
Student-support agent	Partially observable and stochastic because student context may be incomplete and language generation may vary.	May be fully observable and deterministic if all inputs, documents, tool results, and responses are fixed in advance.

Conclusion

Under the stated assumptions, both real systems are partially observable and stochastic. Their simplified simulations may instead be fully observable and deterministic if all relevant information is available and action outcomes are fixed.